AI Exposure and Age-Differentiated Employment: Evidence from Norwegian Register Data

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Abstract

Using Norwegian population administrative data covering 2021 to 2026, we examine age-differentiated employment in occupations differentially exposed to artificial intelligence. We build monthly employment series by occupation, age group, sector, and month, indexed to October 2022, the month before the November 30, 2022 release of ChatGPT, and rank occupations by the Eloundou et al. GPT-4 exposure measure. Selected high-exposure occupations, including software developers and ICT systems analysts, show falling employment among the youngest workers. Across the full exposure distribution, the most-exposed quintile in the private sector does not fall relative to the least-exposed for workers under 40, and a difference-in-differences estimate for this group is small and positive. The negative exposure gradient is concentrated among workers aged 41 to 50, where both employment and new hires in the most-exposed quintile decline relative to the least-exposed after 2022. Public-sector employment shows no exposure gradient. Cash earnings show no systematic divergence by exposure quintile, consistent with adjustment on the hiring margin under Norway's two-tier bargaining system.

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1 Introduction

Is generative AI reshaping employment patterns? A growing literature investigates a post-2022 age gradient in occupations most exposed to artificial intelligence: employment of young workers declines relative to older workers, while no such gradient appears in less-exposed occupations. Brynjolfsson et al. (2025) find this pattern in US payroll data, Lodefalk et al. (2026) document a similar gradient in Sweden, and Klein Teeselink (2025) reports consistent UK evidence. On the other hand, Humlum and Vestergaard (2026) and Kauhanen and Rouvinen (2026) report null patterns from Denmark and Finland.

We document Norwegian employment patterns. Using population-level register data from the Norwegian employer reporting system for 2021 to 2026, we construct monthly employment series at the level of cells defined by occupation, age group, sector, and AI exposure quintile, indexed to October 2022, the month preceding the November 30, 2022 release of ChatGPT. We rank occupations by the Eloundou et al. GPT-4 exposure measure. Selected high-exposure occupations show falling employment among the youngest workers, but across the full exposure distribution the most-exposed quintile does not decline relative to the least-exposed for workers under 40. The negative exposure gradient appears instead among workers aged 41 to 50, and is concentrated in the private sector. Public-sector employment shows no corresponding gradient. Cash earnings within occupation-age cells show no systematic divergence by exposure quintile.

The data cover the entire employed population at monthly frequency, with age assigned at monthly precision. Our main exposure measure is the theoretical task measure of Eloundou et al. (2024); the appendix reports the automation and augmentation shares of Handa et al. (2025), based on revealed Claude usage, as an alternative.

Norway is also a setting where AI tools have diffused rapidly. A 2026 Microsoft report ranks Norway third globally in population-level AI usage, after the UAE and Singapore (Microsoft AI Economy Institute, 2026). OECD data for 2025 place Norway first among 33 European countries for AI use at work (Ostertag, 2026). At the enterprise level, the share of Norwegian firms (10 or more employees) reporting AI use rose from 11 percent in 2021 to 30 percent in 2025 (Walther-Zhang and Rybalka, 2025), placing Norway well above the EU average of 20 percent; the Nordic countries as a group lead European enterprise adoption (Bick et al., 2026). Internet access (99 percent) and OECD top-five broadband density provide the infrastructure for rapid diffusion (OECD, 2024). If AI-related labor market effects are present, a high-adoption economy is where they would appear first.

2 Related Literature

2.1 Early Empirical Evidence

A small but growing body of work investigates employment shifts associated with generative AI exposure. We review the most closely related studies to ours before turning to null findings and to the measurement of AI exposure.

Brynjolfsson et al. (2025) provide large-scale evidence from the United States. Using private payroll records from ADP covering 3.5 to 5 million workers per month between January 2021 and September 2025, they estimate a Poisson regression event study with firm-by-quintile and firm-by-time fixed effects. Workers aged 22–25 in the most AI-exposed occupational quintile experienced a decline of approximately 15 log points in employment relative to the least-exposed quintile. The effect is concentrated in occupations exposed to automation, as classified by the Handa et al. (2025) decomposition. Occupations exposed primarily to augmentation show no consistent pattern. Wages exhibit little divergence over the same period, consistent with downward nominal rigidity.

Lodefalk et al. (2026) replicate this design in Sweden using employer-employee data covering January 2019 through June 2025. Their difference-in-differences specification includes employer-by-quartile and employer-by-month fixed effects, with AI exposure measured by the DAIOE generative AI index of Engberg et al. (2024) at the four-digit SSYK level. Workers aged 22–25 in the most-exposed quartile show a 5.5 percent employment decline by the first half of 2025. The age gradient is monotonic: the youngest workers decline most, while the oldest (50 and above) gain 1.3 percent. A separate analysis of job postings from Platsbanken attributes part of the decline in vacancies to monetary tightening following the Riksbank rate hike of April 2022, rather than to AI. Pre-trend joint tests reject the null of parallel trends, but Lodefalk et al. argue that the pre-trend slope for the 22–25 group is essentially noise and that any linear extrapolation would bias the estimate toward zero rather than away from it.

Klein Teeselink (2025) uses Revelio Labs data from the United Kingdom and finds that a one standard deviation increase in LLM exposure reduces employment by 0.3 percent. The effects are concentrated in junior positions and in high-wage occupations. Klein Teeselink and Carey (2026) analyze job posting data from 39 countries and find that AI exposure reduces job postings on average. Countries with stricter employment protection exhibit larger hiring reductions, consistent with firms adjusting on the hiring margin when separation costs are high.

Reported magnitudes vary across countries and specifications: Brynjolfsson et al. report a 16 log-point relative decline in the US, Lodefalk et al. a 5.5 percent decline in Sweden, and Teeselink a 0.3 percent decline per standard deviation in the UK. These figures use different units, baselines, and specifications and should not be read as directly comparable magnitudes.

2.2 Null and Mixed Results

Not all studies find displacement. Humlum and Vestergaard (2026) study Denmark using population-wide register data combined with two survey rounds of approximately 25,000 responses each. Their difference-in-differences design compares workers who adopted AI chatbots to non-adopters. They find a precise null on earnings and no employment effects; at the workplace level, total hours at encouraged-use workplaces are precise enough to rule out effects larger than 2 percent. However, adopters who also switched occupations on average moved to occupations with higher AI exposure and experienced faster earnings growth. This represents an increase in full-time-equivalent occupational mobility, transitioning into higher-wage, higher-AI-exposure occupations.

In their Appendix D.2, Humlum and Vestergaard (2026) use the same 22–25 early-career age definition as Brynjolfsson et al. (2025) and document aggregate declines in early-career employment in several of their exposed occupations in Denmark, including software development, customer support, legal professions, and marketing. The Danish aggregate pattern thus mirrors the US pattern. Their firm-level difference-in-differences then shows that these declines are not driven by firms adopting AI chatbots: estimates rule out effects larger than one third of a percentage point on the early-career employment share at adopting workplaces. They suggest that changes in the macroeconomic environment or anticipatory hiring freezes unrelated to actual adoption may be responsible.

Finnish evidence reinforces the null findings. Kauhanen and Rouvinen (2026) use population-level register data and replicate the Brynjolfsson et al. (2025) methodology. They find no systematic displacement effects and attribute observed employment trends to demographic shifts rather than AI exposure. Earlier Finnish analyses by Kauhanen and Rouvinen (2024, 2025) using a C-AIOE exposure measure likewise found null effects.

Measurement, institutional context, and adoption timing differ across these studies. Humlum and Vestergaard (2026) identify adopters through surveys; Brynjolfsson et al. (2025) and Lodefalk et al. (2026) rely on occupation-level exposure indices. If AI displaces through reduced firm entry and exit, individual-level adoption designs may miss the channel. Nordic labor markets share strong employment protection and centralized bargaining that may buffer or delay displacement.

2.3 AI Exposure Measurement

Empirical studies of AI and employment use occupation-level measures that map AI capabilities to task content. The task framework underlying much of this work is set out in (Autor et al., 2003; Acemoglu and Autor, 2011; Acemoglu and Restrepo, 2018, 2019). The measures in current use differ in whether they capture potential task exposure, revealed usage, or historical AI progress on the abilities a job requires.

Eloundou et al. (2024) develop the most widely used framework for LLM exposure. Building on the O*NET task database, they employ human annotators and GPT-4 to rate each task on whether access to an LLM (or LLM-powered software) would reduce completion time by at least 50 percent. The resulting measures distinguish between whether this reduction of at least 50 percent would occur with direct LLM exposure or with complementary software. (E_1) measures the share of an occupation's that are directly exposed, while (E_2) measures the share of tasks exposed through LLM and complementary software. Higher-wage occupations score higher on exposure, reversing the pattern observed with earlier waves of automation. The GPT-4 "beta" score, defined as $E_1 + 0.5 \times E_2$, has become a standard measure in subsequent empirical work.

Handa et al. (2025), or the Anthropic Economic Index, measures exposure through data on actual usage. They use a privacy-preserving classifier to map approximately four million Claude.ai conversations to O*NET tasks, dividing each task's conversation count by the number of occupations it is associated with before aggregating to the occupation level. They decompose usage into automation (directive task delegation and feedback-loop conversations with minimal human involvement) and augmentation (collaborative task iteration, learning, and validation), reporting 43 percent automation and 57 percent augmentation across all conversations. This decomposition is important because the two modes may have opposite labor market implications: automation substitutes for worker effort, while augmentation complements it. AI use is highly concentrated by occupation: only about 4 percent of occupations show usage in at least three-quarters of their tasks and about 36 percent in at least a quarter, with Computer and Mathematical occupations alone accounting for 37 percent of all observed conversations.

Felten et al. (2021) take a different approach, linking AI application benchmarks across ten categories (including language modeling, image recognition, and speech recognition) to the occupational abilities listed in O*NET. Their AI Occupational Exposure (AIOE) score is a weighted sum of AI progress on the abilities required by each occupation. This measure has been extended to a dynamic version (DAIOE) that tracks AI capability growth over time. Engberg et al. (2024) construct a DAIOE variant emphasizing generative AI applications, which is used by Lodefalk et al. (2026) in the Swedish analysis described above.

A deeper conceptual concern is that standard exposure indices aggregate task-level scores using linear formulas, implicitly treating tasks as separable. Gans and Goldfarb (2025) argue that in many occupations, tasks are quality complements rather than independent: output is multiplicative across tasks, as in an O-ring production function (Kremer, 1993). Under task complementarity, automating some tasks frees the worker to focus on the remaining ones, raising their quality and potentially increasing labor income. The relevant object is then not average task exposure but the structure of bottleneck tasks and how automation reshapes worker time around them. Imas and Shukla (2026) extend this logic, arguing that job dimensionality (the number of distinct tasks in a job) determines whether AI augments or displaces. Low-dimensionality jobs built around a small number of core tasks face higher displacement risk even if their average exposure score is low, while high-dimensionality jobs with many complementary tasks are more likely to be augmented. Exposure indices that average across tasks will overstate displacement risk for complex jobs and understate it for narrow ones.

For Norway specifically, existing evidence documents growing AI adoption across the economy. Kompetansebehovsutvalget (2026) report that AI-related job postings nearly doubled between 2023 and 2024, concentrated in IT but spreading to other sectors.

3 Data, Measures, and Approach

3.1 Norwegian Labor Market Data

Our labor market data come from the *A-ordningen*, Norway's coordinated employer reporting system. Every Norwegian employer submits a monthly report of all employment relationships, wages, and tax withholdings. The Norwegian Tax Administration administers the system on behalf of Statistics Norway and the Norwegian Labour and Welfare Administration. Statistics Norway compiles the reports into the ARBLONN register, which covers the universe of formal employment in Norway except self-employment, approximately 3.1 million employment records per month (Statistics Norway, n.d.a). We access the data through microdata.no, Statistics Norway's remote analysis platform, which provides programmatic access to population-level registers and exports aggregated cell-level tables.

Reference period and inclusion rules. ARBLONN is a monthly status register. The reference period is the week containing the 16th of each month, almost always week 3

(Statistics Norway, n.d.b). On microdata.no, every monthly observation is stamped to the 16th (`YYYY-MM-16`). An employment relationship is included in month t if its start date is on or before the last day of the reference week and either no end date is recorded or the end date is on or after the first day of the reference week. Two further rules apply. First, when cash earnings are reported for the calendar month but cannot be linked to a contemporaneous or prior-month employment relationship, Statistics Norway constructs a fictional employment relationship and classifies the worker as employed. Second, when wage is paid in the month after an employment relationship ends, the relationship is retained for the last month in which it covered the reference week. As a result, monthly counts cover slightly more workers than a strict snapshot on the 16th would.

Variables. The unit of observation is a person–firm pair in a given month. Statistics Norway aggregates wage components reported by the same employer for the same person into a single person–firm record (Statistics Norway, n.d.b). Table 1 lists the variables we use; all are measured at the 16th of the month.

Table 1: ARBLONN variables used in the analysis

Variable	Description
Occupation code	Four-digit STYRK-08 code for the employment relationship; identical to the International Standard Classification of Occupations (ISCO-08) at the four-digit level (Statistics Norway, 2011).
Cash earnings	Total cash compensation paid by the employer during the calendar month, gross of tax; includes base salary, fixed and irregular supplements, bonuses, overtime pay, and severance.
Hourly wage	Hourly wage reported by the employer.
Contracted position percentage	Contracted hours as a share of full-time, with 100 representing full-time employment.
Overtime hours	Overtime hours reported by the employer for the calendar month.
Start date	Contractual start date of the employment relationship active on the 16th. We define a new-job indicator equal to one if the start date falls within the 30 days ending on the 16th of month t.
Institutional sector	Four-digit code from Statistics Norway's Business Register, collapsed into state (codes 1110, 1120, 6100), municipal (codes 1510, 1520, 6500), and private (all other codes).

Sample. Age is computed at monthly precision by subtracting birth year from calendar year and decrementing by one if the birth month exceeds the current month. Workers are assigned to four decade age groups: 21–30 (early career), 31–40, 41–50, and 51–60 (senior). We exclude individuals under age 21, whose labor market attachment is dominated by education transitions, and those over 60, who are affected by retirement timing. The sample period is January 2021 through February 2026.

3.2 Institutional Context

Several features of the Norwegian labor market are relevant for interpreting AI exposure patterns. Norway's public sector employs approximately 30 percent of the workforce, split between state and municipal employers. Wage determination is centralized: unions and employer organizations negotiate sector-level agreements that compress the wage distribution relative to market-based systems such as the United States. Employment protection legislation is strong by international standards, with notice periods and restrictions on dismissal that slow labor market reallocation.

The period we study coincides with a monetary policy tightening: Norges Bank raised the policy rate from zero percent in September 2021 to 4.5 percent by January 2024 (Figure A1). This tightening affected labor demand broadly, and any AI-specific employment patterns need to be read against this macroeconomic background.

A demographic factor also affects the youngest age group. The early-career cohort shrank slightly between 2022-Q1 and 2025-Q1 (Figure A2). Although small, this trend works in the same direction as any AI-related decline in young employment, and should be kept in mind when reading the indexed series.

A small set of Norwegian reports document AI adoption and its early labor-market correlates. Jordell et al. (2026) surveys 4,294 firms and reports that the share using AI rises from 24 percent in 2023 to 55 percent in 2025, concentrated in ICT, finance, and professional services. Vatne et al. (2026) regresses occupation-level unemployment growth on a Claude-based exposure measure for 2022–2025 and reports a positive gradient and no wage relationship, without decomposition by age or sector. Flobakk-Sitter et al. (2024) interviews members of public-sector unions and finds AI use that is exploratory and concentrated in augmentation rather than automation. Riksrevisjonen (2024) reports that more than half of Norwegian state agencies have no experience with AI tools, and Arbeids- og velferdsdirektoratet (2025) reviews the broader labor-market outlook without occupation-level adoption data.

3.3 AI Exposure Measures

Our main measure of occupational AI exposure is the Eloundou et al. (2024) GPT-4 exposure score. As a complement, the appendix uses the Handa et al. (2025) automation and augmentation shares, which capture revealed patterns of AI usage. Table 3 summarizes these measures.

Eloundou et al. (2024), GPT-4 exposure. Eloundou et al. (2024) combine expert and GPT-4 assessments of each occupation's tasks to estimate the share of tasks for which access to a large language model (LLM) would reduce completion time by at least 50 percent. The occupation-level score, denoted β, is defined as $E_1 + 0.5 \times E_2$, where E_1 is the share of tasks directly exposed to an LLM and E_2 is the share exposed when complementary software is included. The measure captures theoretical LLM capability as assessed in early 2023. We map 397 STYRK-08 codes, covering 97.5 percent of all four-digit occupations in our data.

Handa et al. (2025), Anthropic Economic Index. Handa et al. (2025) measure revealed AI usage patterns from a sample of Claude.ai conversations, with each conversation classified by O*NET task. The occupation-level score reflects the share of conversations associated with that occupation's tasks, decomposed into automation and augmentation modes as described in Section 2. We map 352 STYRK-08 codes (86.5 percent coverage). The lower coverage reflects the concentration of observed Claude usage: the top seven task categories account for 22 percent of all conversations, leaving many occupations with negligible observed usage.

Crosswalk Construction

Both measures originate in US SOC codes. The Handa et al. measure uses SOC 2010 codes directly. The Eloundou et al. measure uses SOC 2018 codes, which we first map to SOC 2010 using the official BLS crosswalk (November 2017). From SOC 2010, we apply the BLS SOC-to-ISCO crosswalk (August 2012, updated June 2015) to obtain ISCO-08 codes. Since STYRK-08 is identical to ISCO-08 at the 4-digit level (Statistics Norway, 2011), this final step is an identity mapping.

The BLS crosswalk contains partial matches: 38.8 percent of its rows are flagged as one-to-many or many-to-one mappings. When multiple SOC codes map to a single STYRK-08 code, we take the unweighted average of their exposure scores. We compute quality flags for each STYRK-08 code, including a partial-match indicator and the maximum fan-out, defined as the number of SOC codes contributing to a single STYRK-08 score. Of the 397 STYRK-08 codes with an Eloundou et al. score, 57.7 percent have at least one partial-match contributor.

3.4 Empirical Approach

We construct normalized time series following the approach in Brynjolfsson et al. (2025), Figures 1–3. Each series is indexed to October 2022 = 1, the month immediately preceding ChatGPT's public release on November 30, 2022. For outcome y in cell c at month t, the normalized series is $\tilde{y}_{c,t} = y_{c,t}/y_{c,\text{Oct 2022}}$. Changes after October 2022 are expressed as proportional deviations from that pre-ChatGPT level.

For each AI exposure measure, we rank all four-digit STYRK-08 occupation codes by their exposure score and assign them to quintiles, from quintile 1 (least exposed) to quintile 5 (most exposed). Quintiles are formed on the occupation distribution (each four-digit code counts once, regardless of its employment share) rather than on the employment-weighted distribution. Quintile assignment is performed separately for each measure, so an occupation may fall in different quintiles under different measures. We then aggregate employment counts and other outcomes within each quintile-by-age-group cell and construct the normalized time series.

The primary outcome is employment indexed to October 2022, by decade age group and exposure quintile. Headline results are for the private sector, where AI-exposed occupations cluster; the public sector is reported in the appendix. Secondary outcomes, available from 2021 onward, include monthly earnings, hourly wages, contracted position percentage, overtime hours, and the share of new jobs.

Data Availability

The AI exposure measures and crosswalk files used in this paper are publicly available. The Eloundou et al. (2024) GPT-4 exposure scores are available at `https://github.com/openai/GPTs-are-GPTs` (the file `data/occ_level.csv`, supplementary data from the *Science* article). The Handa et al. (2025) Anthropic Economic Index and the Anthropic (2026) `job_exposure` measure are available at `https://github.com/anthropics/anthropic-economic-index`. The Felten et al. (2021) AIOE data are available as an appendix to the *Strategic Management Journal* article. The BLS SOC-to-ISCO crosswalk is available at `https://www.bls.gov/soc/soccrosswalks.htm`. The STYRK-08 classification is documented by Statistics Norway at `https://www.ssb.no/klass/klassifikasjoner/7`. Norwegian labor market data are accessed through the microdata.no platform (`https://microdata.no`). The aggregated tabulations underlying our analysis are available from the authors upon request.

Table 2: Five largest occupations by AI-exposure quintile, April 2025.

Code	Occupation	Eloundou score	Employment (Apr 2025)	Share of quintile
Quintile 1 (least exposed)				
5321	Health care assistants	0.113	111,786	22.5%
9112	Cleaners and helpers in offices, hotels and other establishments	0.028	66,847	13.4%
7411	Building and related electricians	0.128	30,705	6.2%
8160	Food and related products machine operators	0.093	25,357	5.1%
5153	Building caretakers	0.000	23,284	4.7%
Quintile 2				
5311	Child care workers	0.249	107,401	22.2%
5329	Personal care workers in health services not elsewhere classified	0.216	74,248	15.4%
7115	Carpenters and joiners	0.144	40,630	8.4%
2342	Early childhood educators	0.230	34,923	7.2%
5120	Cooks	0.193	21,415	4.4%
Quintile 3				
5223	Shop sales assistants	0.342	126,527	16.6%
2341	Primary school teachers	0.329	87,396	11.5%
2223		0.343	65,960	8.7%
4321	Stock clerks	0.325	34,783	4.6%
2221	Nursing professionals	0.343	32,067	4.2%
Quintile 4				
1120	Managing directors and chief executives	0.494	39,090	7.5%
2310	University and higher education teachers	0.486	38,655	7.4%
1420	Retail and wholesale trade managers	0.481	34,154	6.5%
2142	Civil engineers	0.482	27,415	5.2%
1219	Business services and administration managers not elsewhere classified	0.492	26,980	5.2%
Quintile 5 (most exposed)				
2422	Policy administration professionals	0.566	90,947	17.1%
4110	General office clerks	0.595	58,261	10.9%
3322	Commercial sales representatives	0.572	44,166	8.3%
2511	Systems analysts	0.626	32,804	6.2%
2519	Software and applications developers and analysts not elsewhere classified	0.765	22,053	4.1%

Notes: Occupations are ranked by the Eloundou et al. exposure score. Employment is the April 2025 headcount across both sectors and decade age groups 21–60; share is the occupation's percentage of the quintile's total employment. Occupation titles are the official English ISCO-08 labels.

Table 3: AI Exposure Measures Mapped to Norwegian STYRK-08 Occupations

Measure	Type	Vintage	STYRK-08 coverage	Conceptual basis
Eloundou et al. (2024) β	Theoretical	2023	397 / 407 (97.5%)	Expert and GPT-4 assessment of task exposure to LLMs; $\beta = E_1 + 0.5 \times E_2$
Handa et al. (2025) overall	Revealed usage	2025	352 / 407 (86.5%)	Share of Claude conversations per O*NET task
Felten et al. (2021) AIOE	Ability-based	2018–2021	392 / 407 (96.3%)	AI application performance linked to occupational abilities via O*NET
Anthropic (2026) job exposure	Observed, time-weighted	2026	388 / 407 (95.3%)	Time-weighted Claude usage with automation penalty

Notes: All measures are mapped to STYRK-08 via a SOC 2010 → ISCO-08 crosswalk (BLS, 2012); the Eloundou et al. measure additionally uses the BLS SOC 2018 → SOC 2010 crosswalk (November 2017) as a first step. STYRK-08 is identical to ISCO-08 at the four-digit level (SSB Notater 17/2011). When multiple SOC codes map to a single STYRK-08 code, the STYRK-08 score is the unweighted mean of the contributing SOC scores. Coverage is expressed as the share of 407 STYRK-08 four-digit codes observed in our employment data.

4 Results

4.1 Employment in Selected Occupations

Figure 1 displays employment in four occupation groups with high theoretical AI exposure: software developers, ICT systems analysts, customer service agents, and ICT operations technicians.[1] Each panel plots private-sector employment indexed to October 2022 = 1, separately by decade age group, with a vertical line marking the public release of ChatGPT on November 30, 2022.

In software development the 21–30 series falls to about 0.81 by early 2026, while the 31–40 and 51–60 series rise to roughly 1.16 and the 41–50 series is flat. ICT systems analysts and ICT operations technicians show the same shape, with the 21–30 series ending near 0.89 and 0.85. Customer service agents are noisier: the 21–30 series is roughly flat while the older groups end higher.

In all four occupations the youngest workers lose ground relative to older workers, the case-study pattern reported in Brynjolfsson et al. (2025). Section 4.2 incorporates all occupations systematically.

[1] ISCO-08 codes: software developers 2512–2514 and 2519; ICT systems analysts 2511; customer service agents 4222; ICT operations technicians 3511.

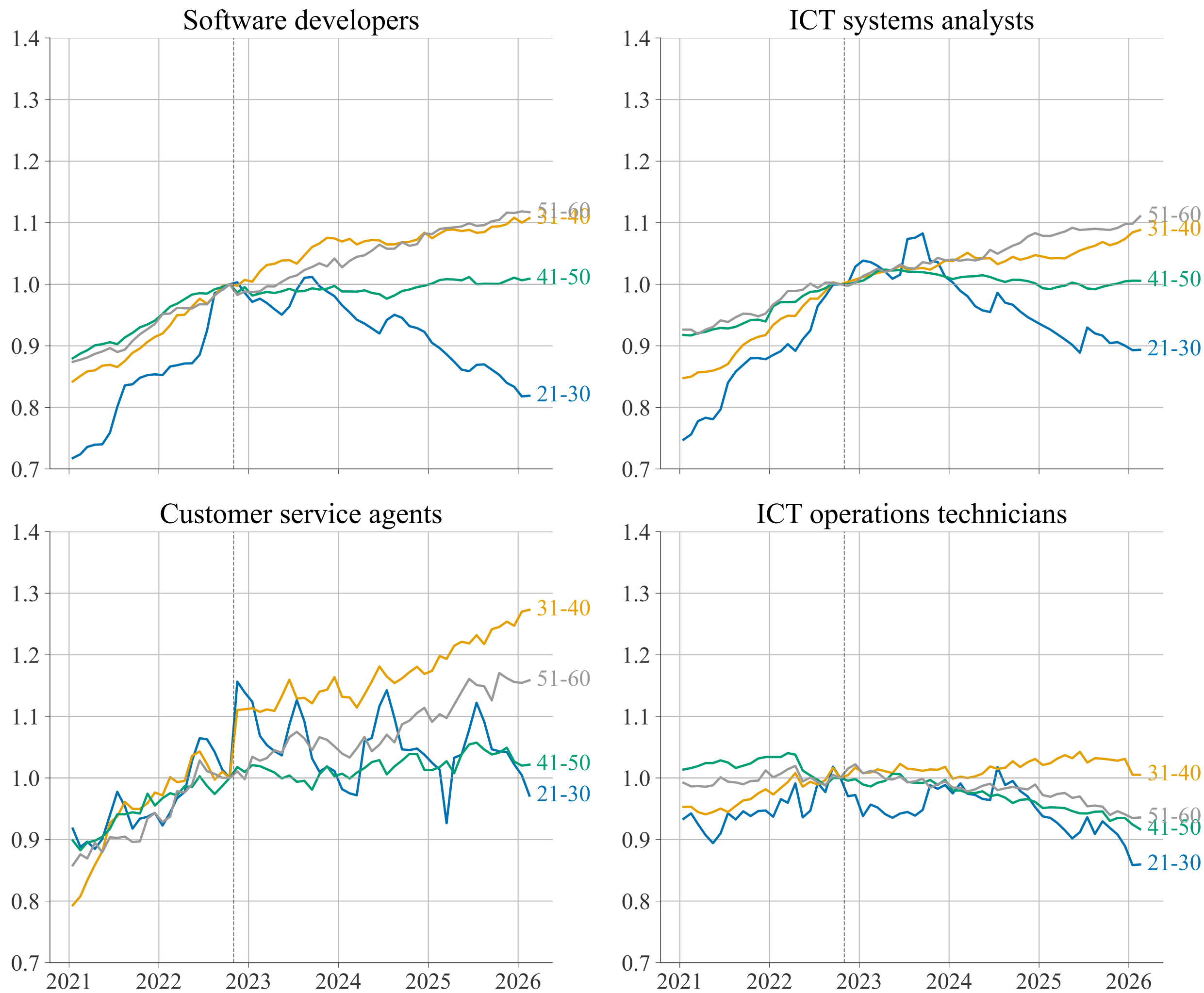

Figure 1: Employment in selected high-exposure occupations, by decade age group.
Notes: Private-sector employment per capita (headcount divided by resident population in the age group), indexed to October 2022 = 1. Panels show software developers (ISCO-08 2512–2514, 2519), ICT systems analysts (2511), customer service agents (4222), and ICT operations technicians (3511). The vertical dashed line marks November 30, 2022 (ChatGPT release).

4.2 Employment by AI Exposure Quintile

The case studies above examine occupations chosen for their AI relevance. A more systematic approach ranks all occupations by their AI exposure score and examines whether high-exposure occupations exhibit different employment trends than low-exposure occupations. We assign each occupation to a quintile based on its Eloundou et al. GPT-4 exposure score, from Q1 (least exposed) to Q5 (most exposed), and plot normalized employment by quintile within each age group.

Eloundou et al. Exposure

Figure 2 plots quintile-by-age employment for the private sector, indexed to October 2022. Each of the four panels is one decade age group, with quintile lines running from light blue (Q1, least exposed) to dark blue (Q5, most exposed) and a red line for the overall age-group trend.

The most-exposed occupations do not lose ground among the youngest workers. In the 21–30 and 31–40 groups the Q5 series tracks or runs above Q1 after the ChatGPT release. The negative gradient appears instead at ages 41–50, where Q5 ends about four points below Q1 by early 2026. The 51–60 group shows no clear separation.

Table 4 reports the corresponding difference-in-differences estimates: the average post-October-2022 change in each quintile relative to Q1, by age group, for employment, new hires, and log monthly earnings. The employment estimates are positive and significant for the most-exposed quintiles at ages 21–40 (Q5 about $+0.07$ log points in both groups) and negative at 41–50 (Q5 -0.053, $p < 0.05$). New hires fall for the most-exposed quintile at 41–50 (Q5 -0.084). Log earnings show no systematic exposure gradient.

The companion automation and augmentation quintile plots based on the Handa et al. decomposition are reported in Appendix Figures A3 and A4.

4.3 Automation versus Augmentation

We follow Brynjolfsson et al. (2025) on the automation-vs-augmentation split. Ranking the youngest workers by Handa automation share gives a modest negative gradient (Appendix Figure A3); ranking them by augmentation share gives no gradient at all (Appendix Figure A4). The split points toward displacement rather than augmentation, but the Handa classification only captures conversation mode, not what workers actually do on the job. An occupation scored as "automation-exposed" through conversation mode may in practice use AI for augmentation.

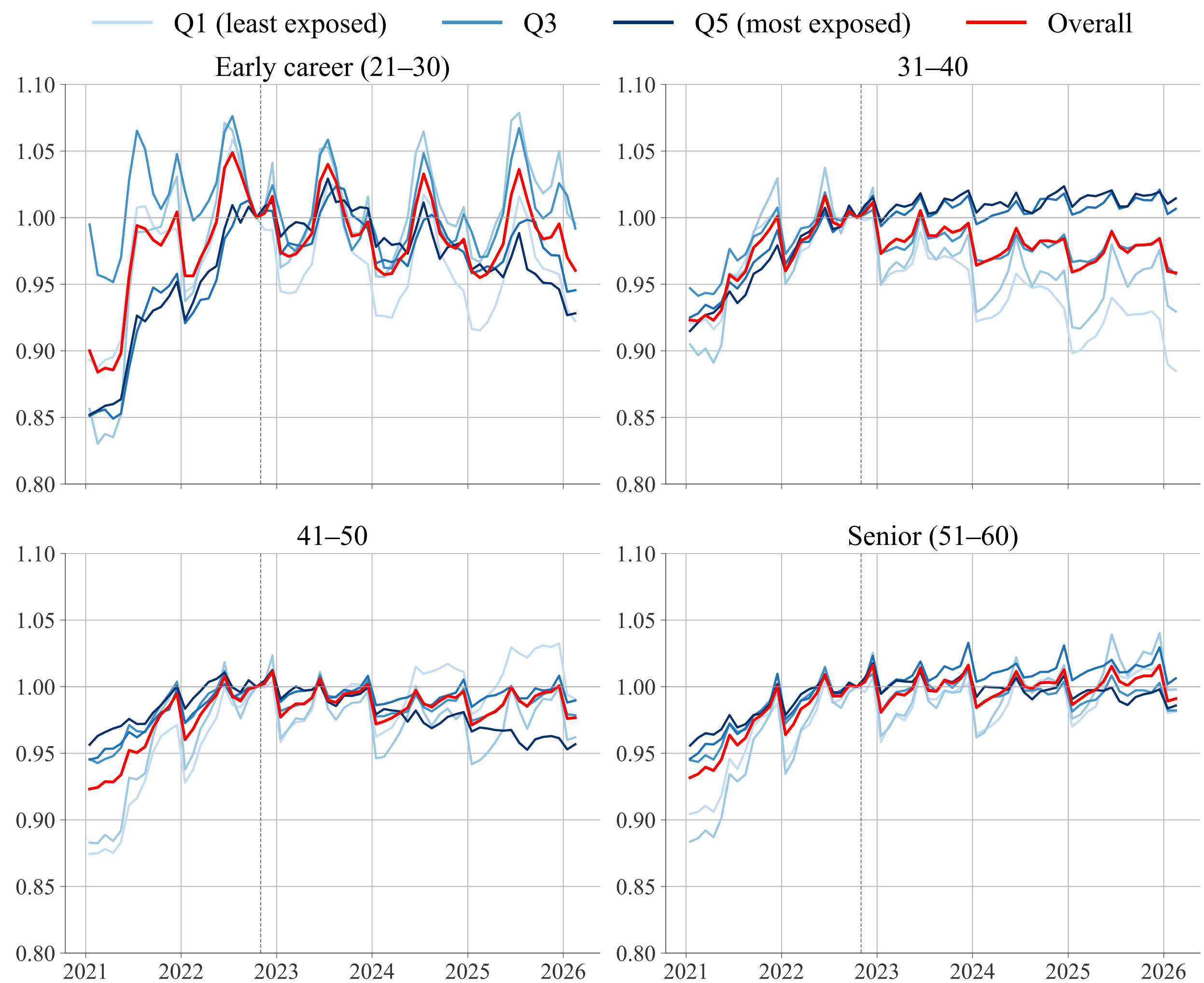

Figure 2: Employment by AI-exposure quintile and decade age group, private sector.
Notes: Private-sector employment per capita (headcount divided by resident population in the age group), indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. GPT-4 exposure score. Q1 (lightest) = least exposed; Q5 (darkest) = most exposed. Red line = overall age-group trend. Vertical dashed line marks the November 2022 ChatGPT release.

Table 4: Cell-level difference-in-differences by AI-exposure quintile, private sector.

	Early career (21–30)	31–40	41–50	Senior (51–60)
	(1)	(2)	(3)	(4)
Panel A. Employment (Poisson)				
Q2 × Post	0.0568	-0.0075	0.0137	0.0233
	(0.0511)	(0.0334)	(0.0302)	(0.0240)
Q3 × Post				
Q4 × Post	0.0774	0.0431**	0.0093	0.0151
	(0.0479)	(0.0183)	(0.0198)	(0.0158)
Q5 × Post	0.0786*	0.0496**	-0.0109	-0.0008
	(0.0455)	(0.0194)	(0.0209)	(0.0198)
Panel B. New hires (Poisson)				
Q2 × Post	0.0091	0.0042	0.0099	0.0129
	(0.0430)	(0.0586)	(0.0636)	(0.0693)
Q3 × Post				
Q4 × Post	-0.0510	-0.0316	-0.0439	-0.0008
	(0.0368)	(0.0386)	(0.0447)	(0.0542)
Q5 × Post	-0.0758*	-0.0879**	-0.1030**	-0.0354
	(0.0393)	(0.0411)	(0.0456)	(0.0540)
Panel C. Log monthly earnings (OLS)				
Q2 × Post	0.0259*	0.0065	0.0022	0.0000
	(0.0141)	(0.0073)	(0.0080)	(0.0060)
Q3 × Post				
Q4 × Post	0.0319**	0.0104*	0.0068	0.0105**
	(0.0141)	(0.0061)	(0.0068)	(0.0052)
Q5 × Post	0.0332**	0.0156**	0.0143**	0.0146***
	(0.0130)	(0.0063)	(0.0067)	(0.0049)
Occupations	393	390	391	391
Observations (count)	20,436	20,280	20,332	20,332
Observations (wage)	19,337	19,439	19,371	19,325

Notes: Each entry is the post-October-2022 change in the indicated quintile relative to Q1 (least exposed), by decade age group. Employment and new hires are Poisson; log monthly earnings is OLS weighted by cell employment. Occupation and month fixed effects; standard errors clustered by occupation. Sample 2021m1–2025m4. *, **, *** denote $p < 0.1, 0.05, 0.01$.

4.4 Sector Heterogeneity

The exposure gradient documented in Figure 2 is a private-sector pattern. The public sector shows no exposure gradient in any age group (Appendix Figure A9).

The public-sector null could reflect slower adoption, hiring insulated from market conditions, stronger employment protection, or a different occupational mix; we cannot separate these from the aggregate data. The adoption channel does have some support: Riksrevisjonen (2024) report that more than half of Norwegian state agencies have no experience with AI tools, so theoretical exposure in the public sector does not translate into implementation. The public sector is dominated by health, education, and administration; private high-exposure occupations cluster in IT, finance, and business services.

4.5 Wages and Other Outcomes

Figure 3 displays monthly cash earnings by Eloundou et al. exposure quintile and age group. Strong seasonal patterns dominate the wage plots: December earnings spike to 1.4–1.5 times the baseline due to Christmas bonuses, and June earnings are elevated due to statutory holiday pay. Brynjolfsson et al. (2025) and Humlum and Vestergaard (2026) find no wage divergence over a similar horizon in the US and Denmark. Norway's two-tier bargaining system (Bhuller et al., 2022) with sectoral wage floors and strong employment protection make the quantity margin the more likely site of adjustment.

Four additional outcomes are reported in the appendix: hourly wages (Figure A6), contracted position percentage (Figure A5), overtime hours (Figure A7), and the share of new jobs (Figure A8). None shows a differential trend by exposure quintile, consistent with the extensive margin as the candidate site of adjustment.

4.6 Poisson Regression Estimates

The quintile series so far normalize employment by the population in each age group, the right denominator for asking whether cohort-level employment shifts away from AI-exposed occupations. We turn now to the worker counts and ask if they reallocate across occupations? Following Brynjolfsson et al. (2025), we estimate Poisson event studies on the counts.

The cell-level microdata are tabulated at the (occupation $\times$ age group $\times$ month) level. We estimate

$$\log E[\text{count}_{j,a,t}] = \alpha_j + \beta_t + \gamma_{q,k},$$

separately per decade age group a in the private sector, where j indexes occupations and

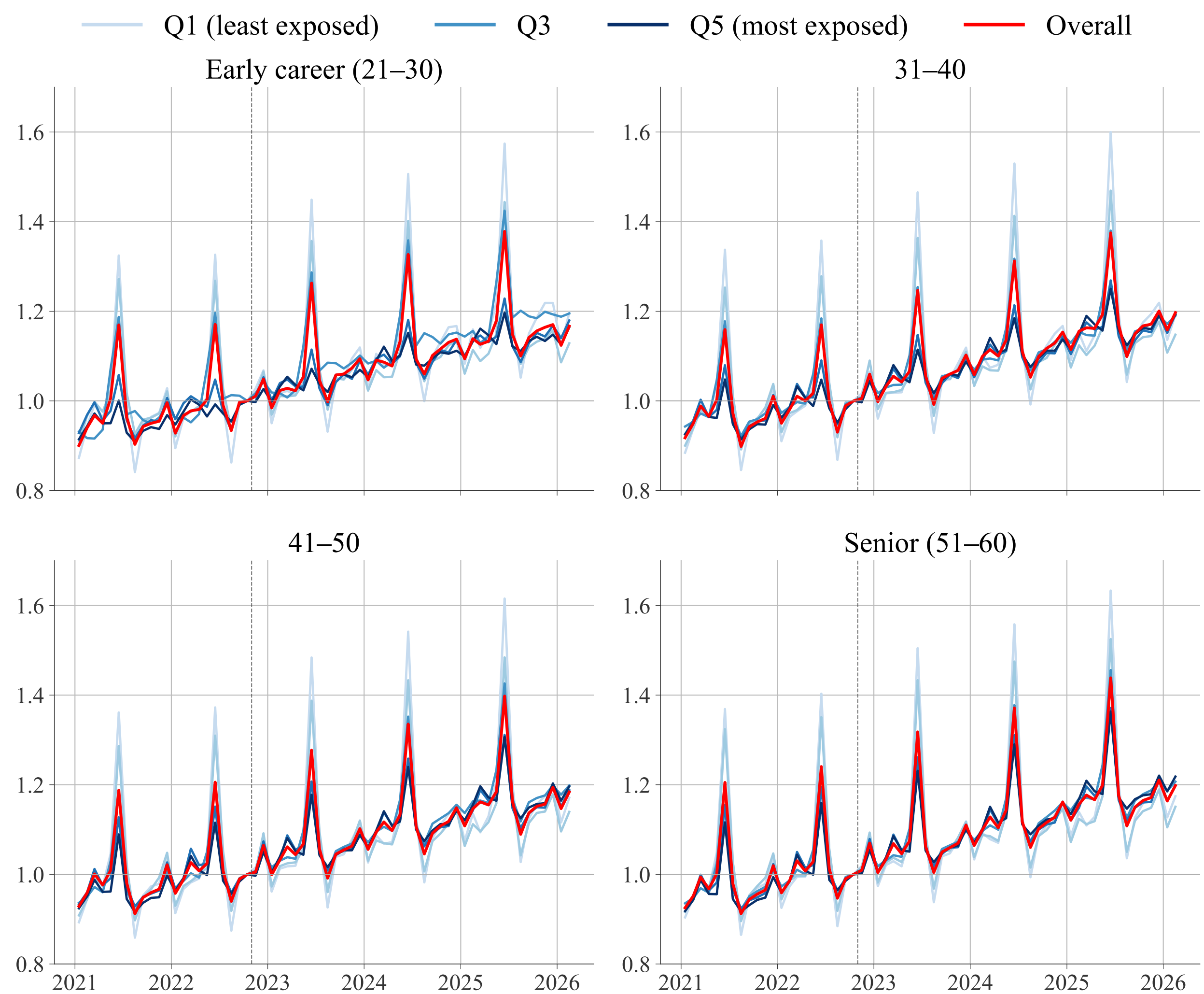

Figure 3: Monthly cash earnings by AI-exposure quintile and decade age group, private sector.

Notes: Mean monthly cash earnings, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score. Strong seasonal patterns (December bonuses, June holiday pay) are visible in all panels.

$q(j)$ is the Eloundou GPT-4 β quintile of occupation j. The occupation FE absorbs the permanent employment level of each occupation; the month FE absorbs the aggregate time path of the age cohort. Standard errors are clustered at occupation. Reference: Q1 and $k = -1$ (October 2022).

Figure 4 reports the resulting $\gamma_{q,k}$ for $q \in \{2, 3, 4, 5\}$ across the four decade age groups, in log points. The path matches the indexed series in Figure 2. At 41–50 the Q5 path turns negative after October 2022 and reaches about -0.03 log points by early 2026. At 21–30 the Q5 path is noisy and ends slightly positive; at 31–40 it rises to about $+0.11$; at 51–60 it stays near zero.

The cell-level estimates absorb the average employment path of each occupation but do not separate reallocation *within* firms from changes in the firm composition. Using the individual-level records on Statistics Norway's secure server, we estimate the within-firm composition design of Brynjolfsson et al. (2025),

$$\log E[\text{count}_{f,q,a,t}] = \alpha_{f,q} + \beta_{f,t} + \gamma_{q,k},$$

on the panel of firm f $\times$ quintile q $\times$ month t cells, separately per decade age group. Firm-by-quintile and firm-by-month fixed effects absorb the firm's persistent composition and any firm-level demand shock; identification rests on within-firm reallocation across exposure quintiles. Standard errors are clustered at the firm. The sample is private-sector *foretak* with at least 20 workers in the 22–55 age window per month, January 2021 through July 2025 (the secure-server window is shorter than the cell-level data by seven months).

Figure 5 reports the firm-FE $\gamma_{q,k}$. The qualitative pattern lines up with the cell-level estimates: at 41–50 the Q5 path drifts negative after late 2022, reaching about -0.05 log points by mid-2025; at 31–40 it rises to about $+0.10$; at 21–30 it is noisy with a sharp upward swing at the end of the window; at 51–60 it stays near zero. The within-firm comparison therefore does not reverse the cell-level pattern, suggesting that the post-2022 reallocation we document operates inside firms rather than only across them.

5 Robustness

5.1 Population Adjustment

The main employment figures report employment per capita, dividing headcount by the resident population in each decade age group (Statistics Norway table 07459, interpolated from annual to monthly), which removes the mechanical effect of changing cohort sizes.

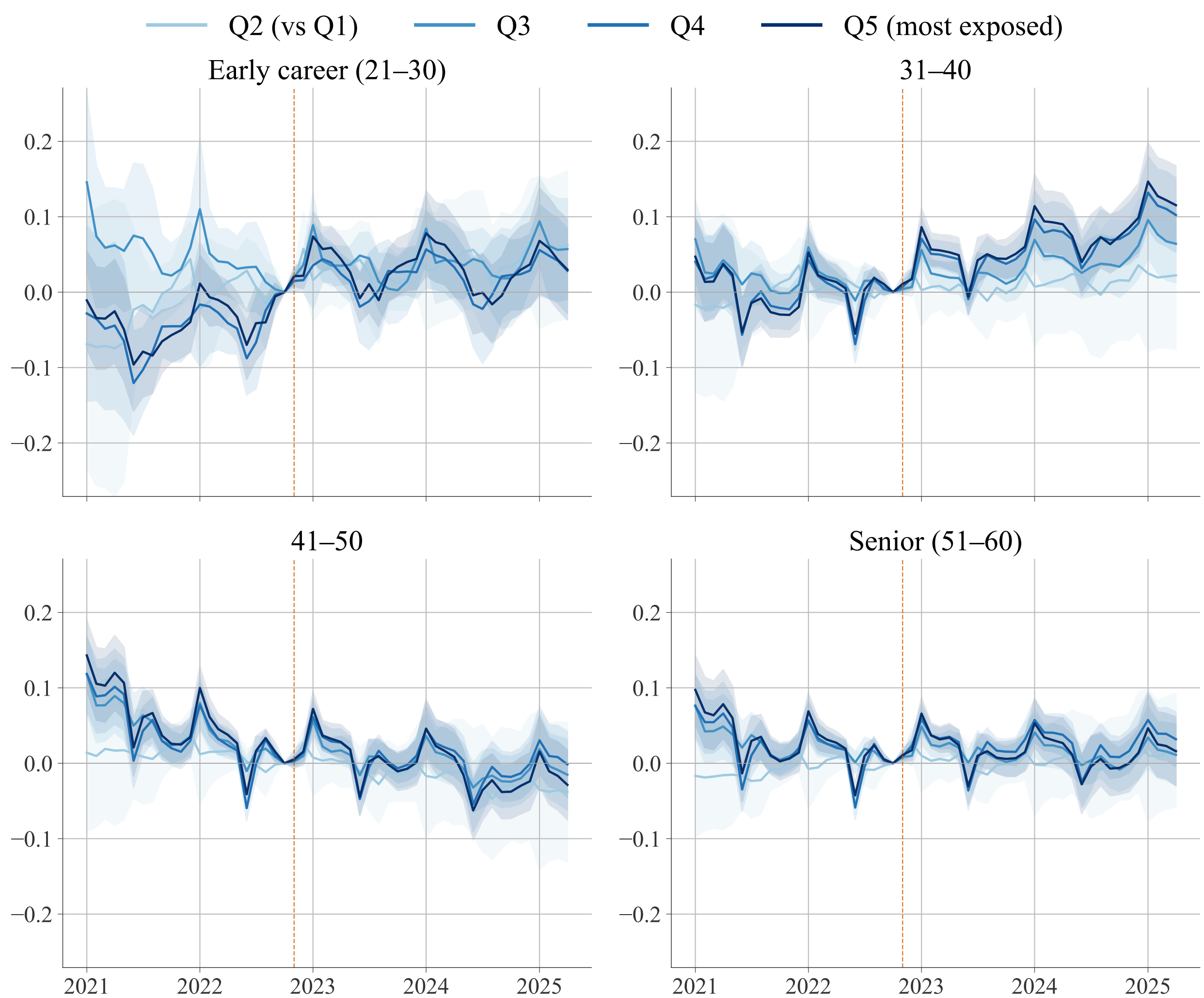

Figure 4: Poisson event-study coefficients by AI-exposure quintile and decade age group, private sector.

Notes: Coefficients $\gamma_{q,k}$ for $q \in \{2, \ldots, 5\}$ relative to $q = 1$, in log points. Cell-level microdata.no aggregates (occupation × age × month), private sector; fixed effects for occupation and month. Sample 2021m1–2025m4. Shaded bands: 95% confidence intervals clustered at occupation. Reference: October 2022. Dashed line marks the November 2022 ChatGPT release.

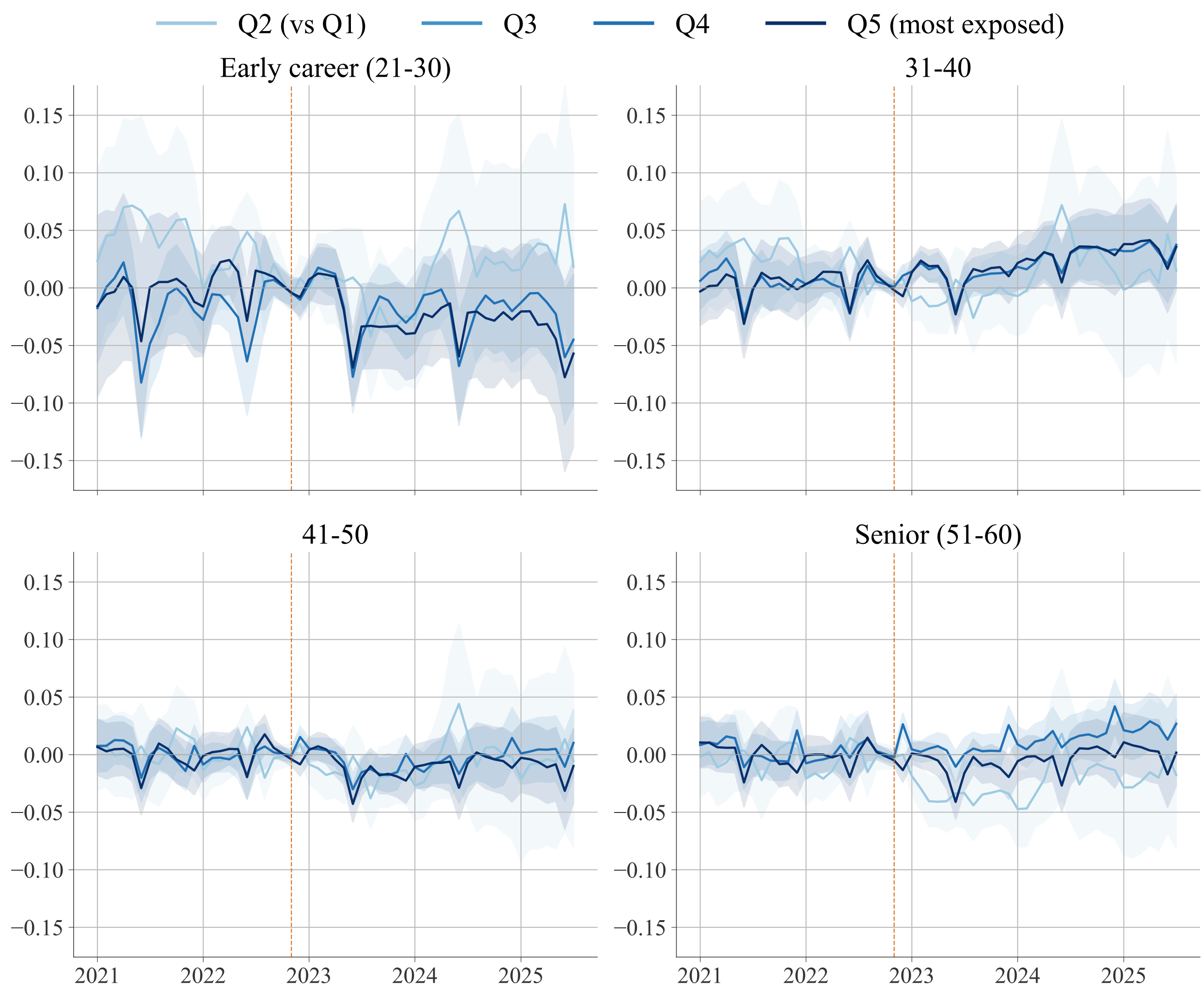

Figure 5: Firm-fixed-effects Poisson event-study coefficients by AI-exposure quintile and decade age group, private sector.

Notes: Coefficients $\gamma_{q,k}$ for $q \in \{2, \ldots, 5\}$ relative to $q = 1$, in log points. Estimated on the individual-level firm $\times$ quintile $\times$ month panel; fixed effects for firm $\times$ quintile and firm $\times$ month. Private-sector foretak with at least 20 workers in the 22–55 age window per month. Sample 2021m1–2025m7. Shaded bands: 95% confidence intervals clustered at the firm. Reference: October 2022. Dashed line marks the November 2022 ChatGPT release.

Population in the decade groups changes slowly over the sample, so the per-capita series differ little from the raw-count series on the same base.

Additional caveats. The Norwegian occupation coding is stable over 2021–2026 by design, but as AI-relevant occupations evolve, employers may shift how they classify ambiguous job titles, introducing a reclassification channel that runs in the opposite direction from the one we are after.

6 Discussion

6.1 Cross-Country Comparison

Table 5 situates the Norwegian patterns within the cross-country evidence reviewed in Section 2. The table separates descriptive time series (Panel A) from formal econometric estimates (Panel B), because several studies employ both approaches and the magnitudes often differ.

All five countries with descriptive analyses show an early-career decline in AI-exposed occupations, with the exception of Finland. The Norwegian per-capita decline of approximately 30 percent for young software developers is between the US (−20%, Brynjolfsson et al., 2025) and the Swedish raw figure (−44%, Lodefalk et al., 2026). The Finnish null (Kauhanen and Rouvinen, 2026) is attributed to demographics.

When formal methods control for firm-level confounders, the magnitudes tend to be smaller. Brynjolfsson et al. (2025) find −16 log points in a Poisson event study with firm fixed effects, compared with larger raw occupation-level declines. Lodefalk et al. (2026) find −5.5% within employer by 2025H1, and their job-posting analysis suggests that much of the aggregate decline in AI-exposed occupations reflects the Riksbank rate cycle. Humlum and Vestergaard (2026) document an aggregate Danish decline in early-career employment that resembles the US and Norwegian patterns, but their firm-level difference-in-differences, the only study with individual-level AI adoption data, does not find that workplace AI chatbot adoption drives the decline, bounding the earnings effect at less than 2 to 3 percent.

The gap between descriptive and formal estimates need not imply that AI plays no role. Formal methods absorb variation that may include both confounders and genuine AI effects operating through channels other than direct firm-level adoption (for example, competitive pressure from AI-adopting rivals, or supply-side reallocation by workers anticipating AI disruption). Nonetheless, the pattern suggests that aggregate descriptive trends, including those we report for Norway, should be interpreted with caution. Part of the observed decline may reflect concurrent forces such as monetary tightening, post-COVID corrections, or seniority dynamics rather than AI-specific adjustment. At the industry level, Bick et al. (2026) find no clear evidence that AI adoption is associated with employment changes in either Europe or the US, consistent with the wage and employment nulls in Humlum and Vestergaard (2026). Norway, Sweden, Denmark, and Finland share similar labor market institutions, yet the formal results differ across countries, so institutional structure alone does not explain the heterogeneity.

Table 5: Cross-Country Comparison of Early Labor Market Patterns under Generative AI

Country	Study	Method	Direction (22–25)	Notes
Panel A: Descriptive time series				
US	Brynjolfsson et al. (2025)	Descriptive	Decline	SW developers 22–25: -20%; top-2 quintiles: -6%
Sweden	Lodefalk et al. (2026)	Descriptive	Decline	$\approx -44\%$ raw in top quartile (Fig. A23)
Denmark	Humlum & Vestergaard (2026)	Descriptive	Decline	Aggregate early-career decline mirrors US (App. D.2)
Finland	Kauhanen & Rouvinen (2026)	Descriptive	Null	Trends reflect demographics, not AI
UK	Teeselink (2025)	Descriptive	Decline	High-wage, junior positions
Panel B: Formal estimation				
US	Brynjolfsson et al. (2025)	Poisson event study, firm FE	Decline	-16 log pts, most vs. least exposed quintile
Sweden	Lodefalk et al. (2026)	DiD, employer FE	Decline	-5.5% within-employer by 2025H1; posting decline driven by rate hikes, not AI
Denmark	Humlum & Vestergaard (2026)	DiD, worker/workplace FE	Null	Aggregate decline not driven by firm AI adoption; rules out $> 2\%$ earnings effect
UK	Teeselink (2025)	DiD	Decline	Junior positions at high-wage firms

Notes: Studies sorted by year within each panel. "Direction" refers to the qualitative sign of the relative employment change for workers aged 22–25 in the most AI-exposed occupations after November 2022. Panel A reports descriptive patterns (raw or per-capita trends); Panel B reports results from formal econometric methods with controls and fixed effects. The descriptive-formal gap is substantial in all studies that employ both: Lodefalk et al. find -44% raw vs. -5.5% in the employer-FE event study; Humlum & Vestergaard find an aggregate decline descriptively but a precise null when using firm-level adoption as treatment. Magnitudes across rows use different units and baselines and are not directly comparable.

6.2 Alternative Explanations

Four non-AI channels could produce the same age-by-exposure gradient, and the aggregate data cannot fully separate them.

Post-COVID technology correction. Software, data analysis, and digital services hired aggressively from 2020 through 2022, disproportionately absorbing young workers. The subsequent correction, as pandemic-era demand for digital services normalized, would reproduce the post-2022 pattern with no AI involvement. The 22–25 software developer line in Section 4 starts to slope down before ChatGPT's release, consistent with this channel.

Monetary tightening. The Norges Bank rate cycle (Section 3.2) overlaps the post-ChatGPT window. Higher rates depress labor demand in finance, real estate, construction, and technology, sectors that also overlap substantially with the high-exposure quintiles, and that shed junior workers first. The public-sector null fits a market-driven story but cannot separate AI from rates at the aggregate level.

Seniority dynamics. In any downturn, employers in technology-intensive sectors retain experienced workers and cut entry-level hiring, producing a mechanical age gradient. Separating this from AI-driven substitution requires within-firm variation, which is exactly what the firm-FE design in Section 4.6 provides. There, the within-firm Q5-vs-Q1 coefficient for 22–25 is near zero, so the cohort-level Q5 decline runs through which firms are hiring rather than which workers existing firms retain. That pattern fits hiring freezes better than internal substitution.

Supply-side reallocation. Workers entering the labor market after 2022 may be avoiding occupations they perceive as exposed to AI. This would generate the same young-worker pattern in exposed occupations through worker choice rather than employer substitution.

The exposure gradient in Figure 2 is a relative pattern: in the 41–50 group the most-exposed quintile declines relative to the least-exposed, while the overall age-group line is roughly flat. The aggregate data alone cannot distinguish occupational reallocation from outright exit. The channels above and the AI channel can all be operating at once.

6.3 Implications

The public-sector null and the wage null are consistent with Norway's two-tier bargaining system (Bhuller et al., 2022) channelling any AI-related adjustment toward the hiring margin in the private sector. Downward-rigid wage floors limit short-run wage adjustment; the locally negotiated drift channel could deliver a wage response over a longer horizon. The field has not converged on a standard "AI exposure" definition, and the results should be read with that measurement uncertainty in mind.

7 Conclusion

This paper examines a post-2022 age-exposure gradient in Norwegian employment using aggregated cell counts from population administrative data (2021–2026). In the private sector, the most-exposed quintile declines relative to the least-exposed among workers aged 41 to 50 rather than among the youngest workers, and there is no corresponding wage divergence. Public-sector employment shows no gradient.

AI-related adjustment, the post-COVID correction in technology hiring, monetary tightening by Norges Bank, and seniority-based retention could each generate the same age-exposure gradient, and the aggregate data we have cannot tell them apart. Humlum and Vestergaard (2026) document the same aggregate pattern in Denmark and show that observable workplace adoption of AI chatbots accounts for at most a small fraction of it. Worker- and firm-level designs in Brynjolfsson et al. (2025), Lodefalk et al. (2026), and Humlum and Vestergaard (2026) answer related questions at finer units of observation.

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Appendix A: Additional Figures

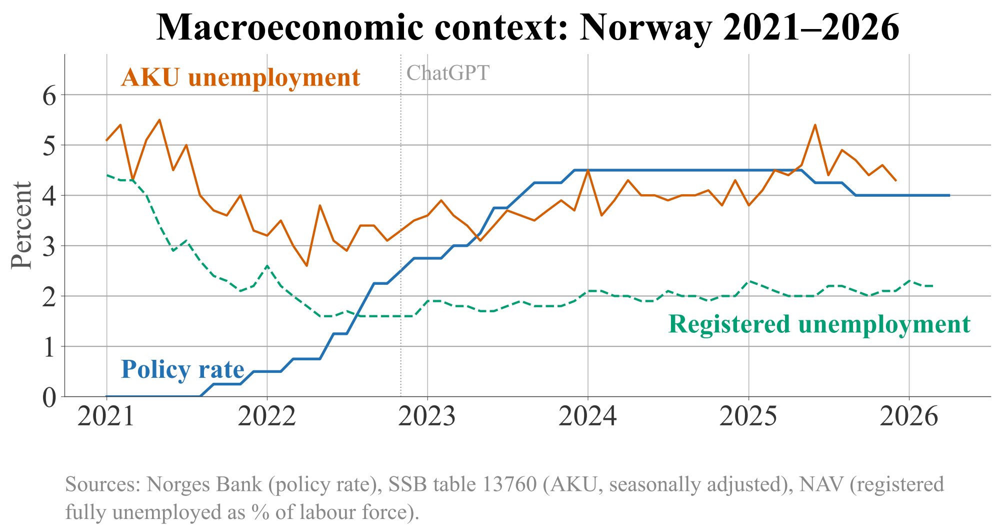

Figure A1: Macroeconomic context: Norges Bank policy rate and registered unemployment rate, 2015–2025.

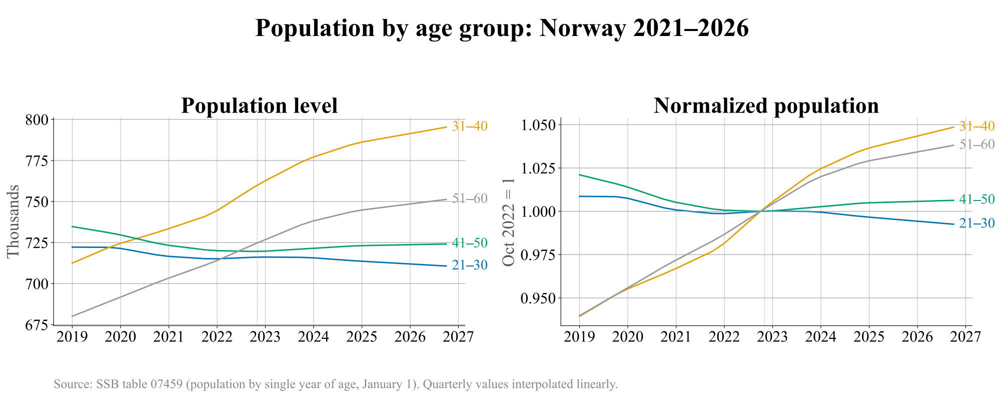

Figure A2: Resident population by decade age group, 2015–2025.

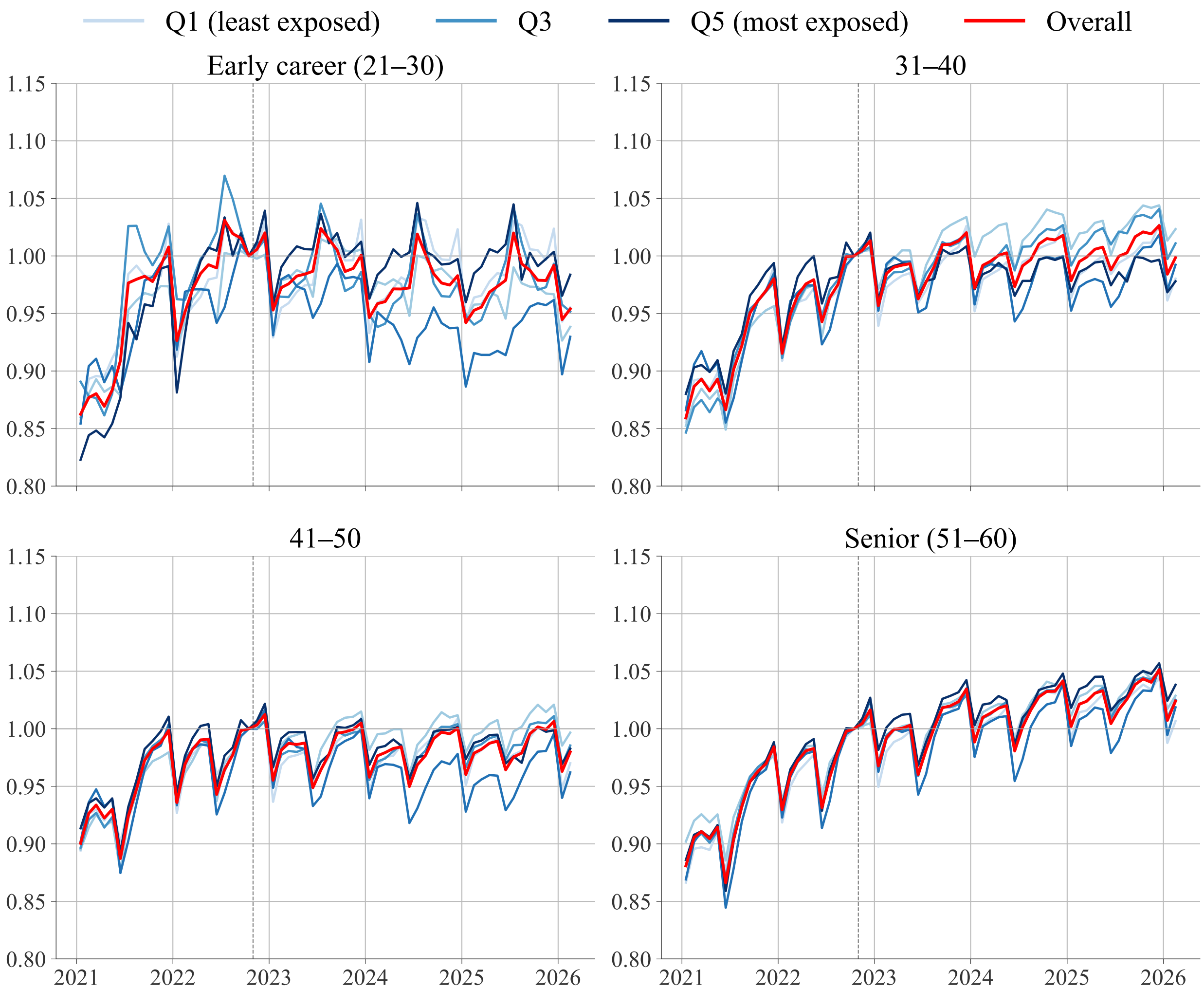

Figure A3: Employment by Handa et al. automation-exposure quintile and decade age group, private sector.

Notes: Private-sector employment indexed to October 2022 = 1. Occupations are ranked by the Handa et al. automation share.

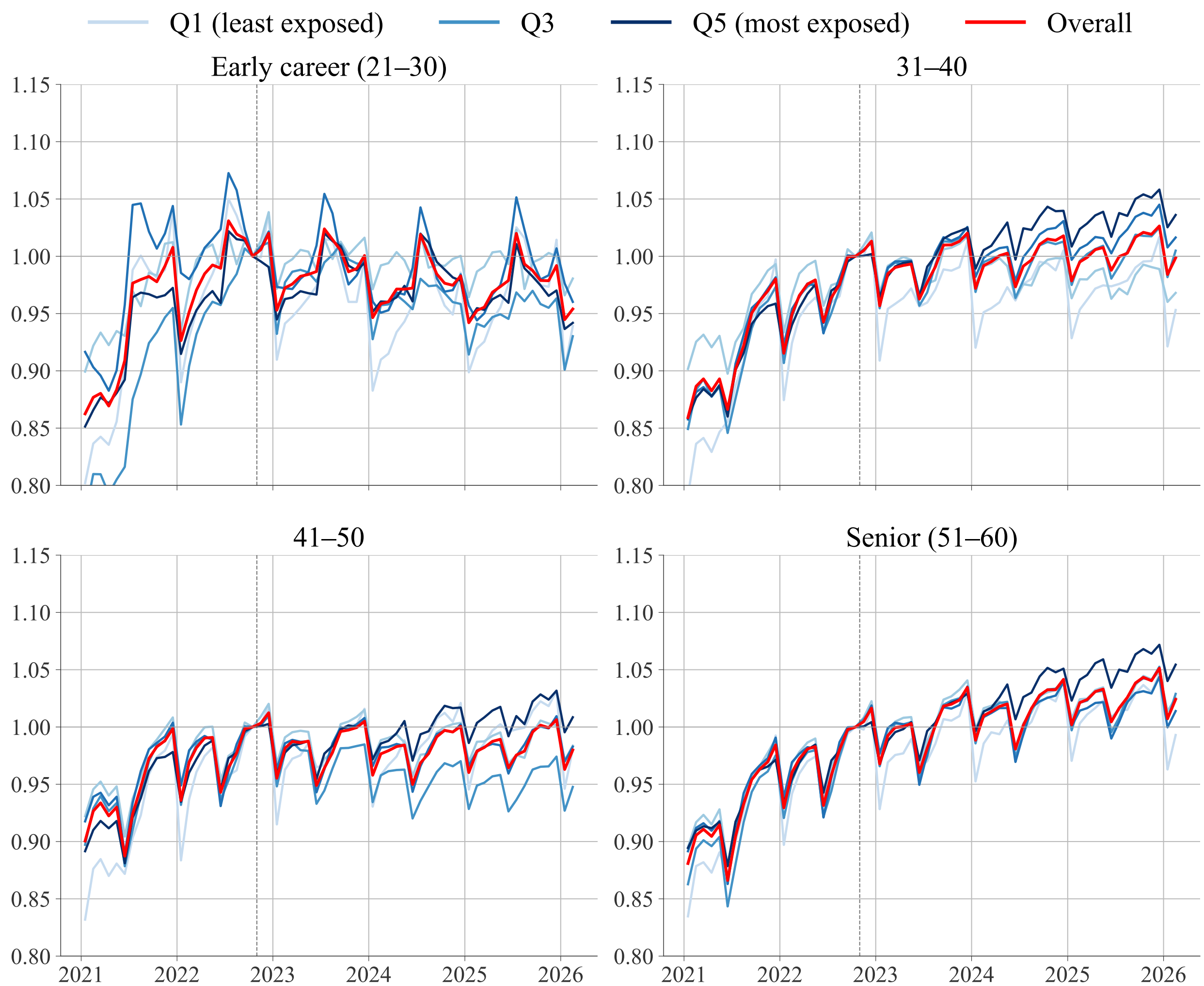

Figure A4: Employment by Handa et al. augmentation-exposure quintile and decade age group, private sector.

Notes: Private-sector employment indexed to October 2022 = 1. Occupations are ranked by the Handa et al. augmentation share.

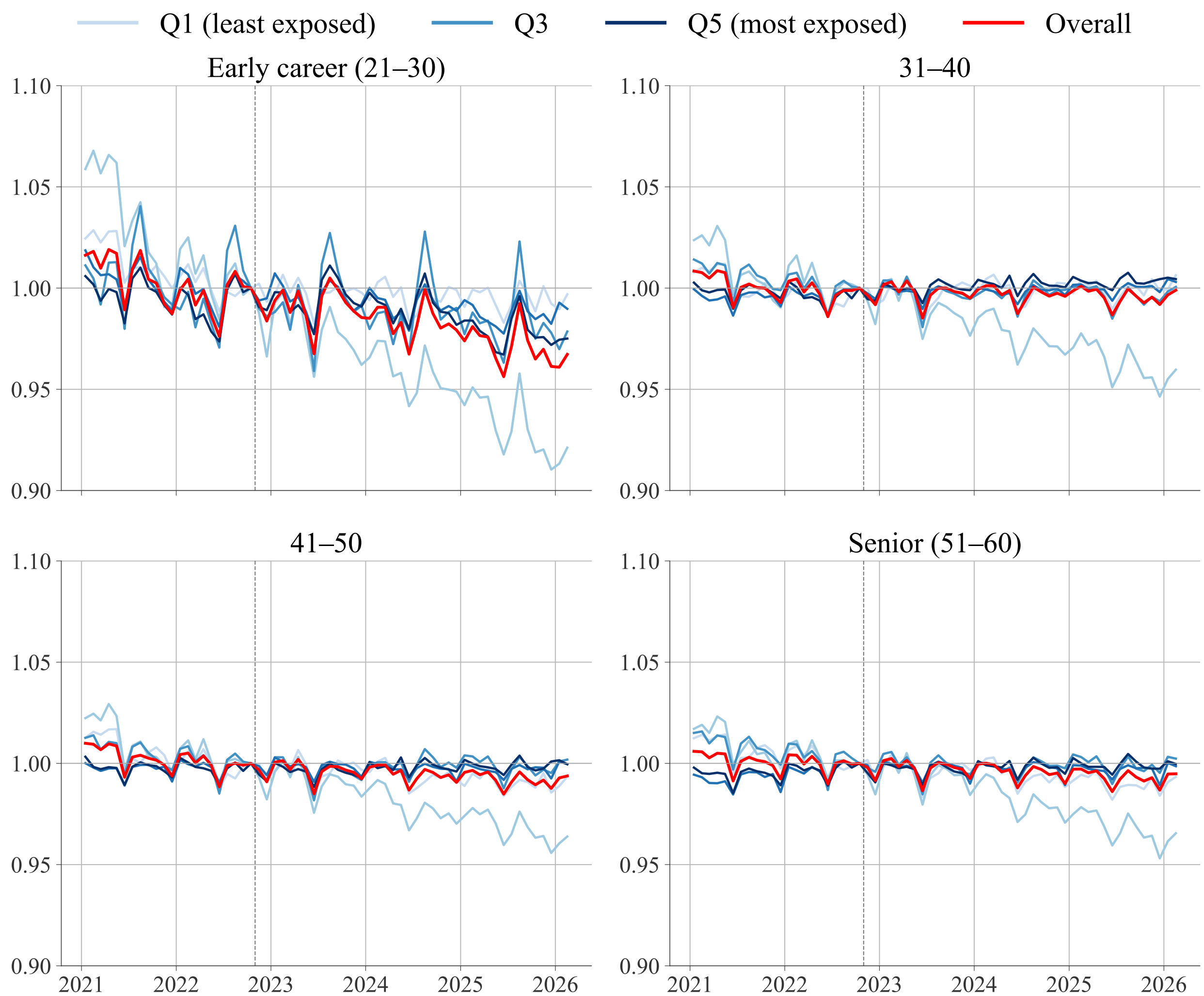

Figure A5: Position percentage by AI-exposure quintile and decade age group, private sector.

Notes: Mean contracted position percentage, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score.

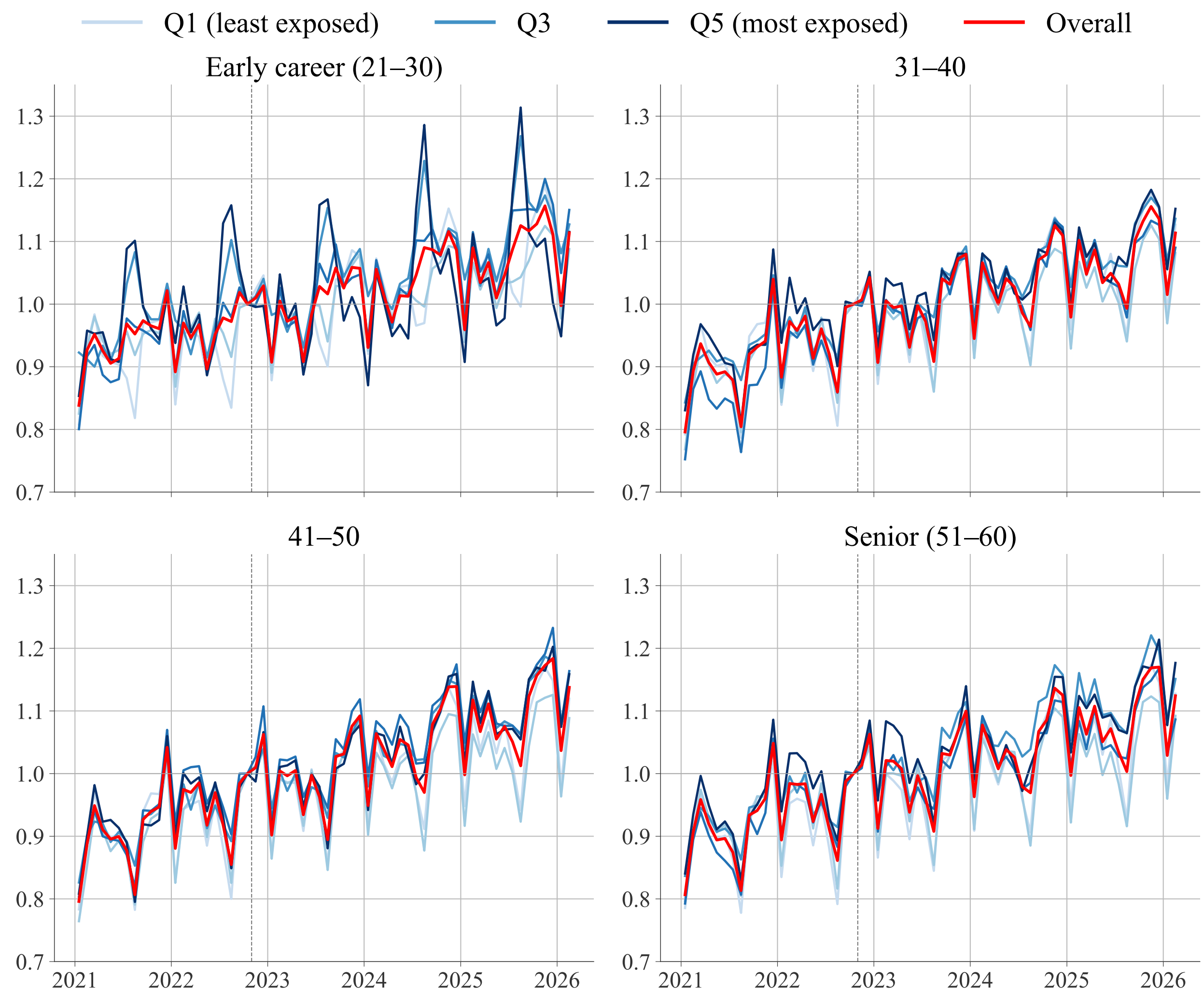

Figure A6: Hourly wage by AI-exposure quintile and decade age group, private sector.
Notes: Mean hourly wage, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score.

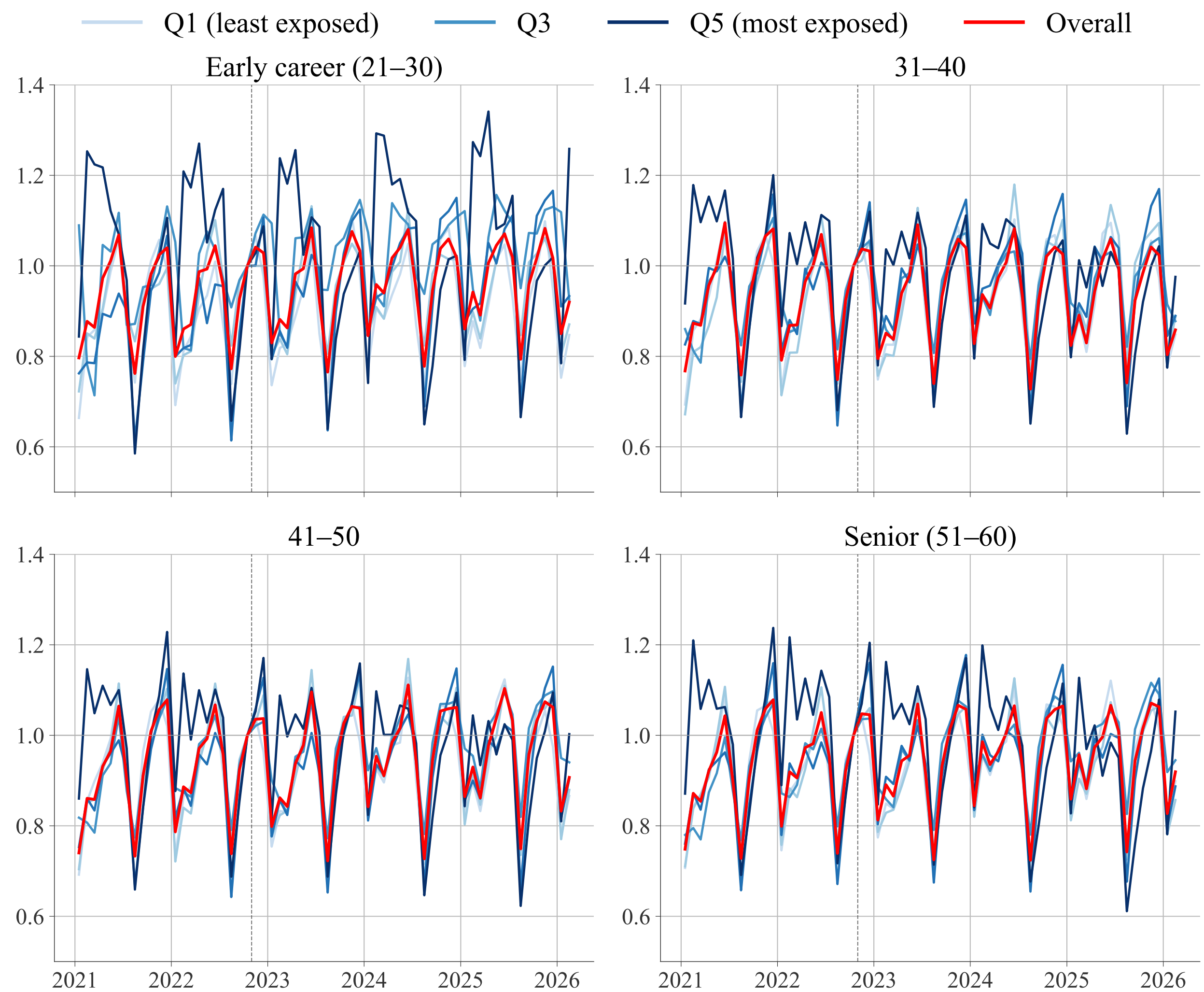

Figure A7: Overtime hours by AI-exposure quintile and decade age group, private sector.
Notes: Mean overtime hours, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score.

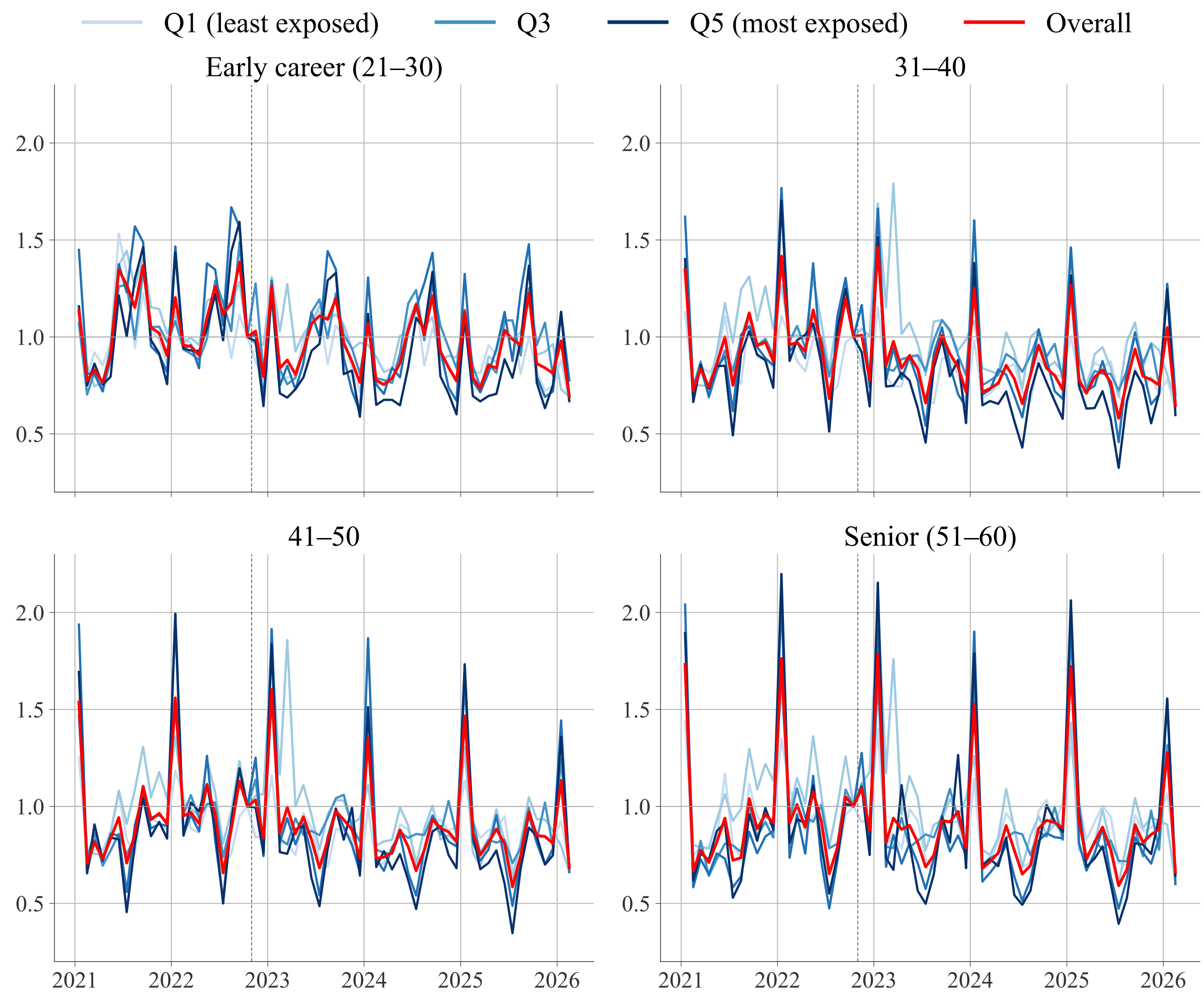

Figure A8: New-hire share by AI-exposure quintile and decade age group, private sector.
Notes: Mean new-hire share, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score.

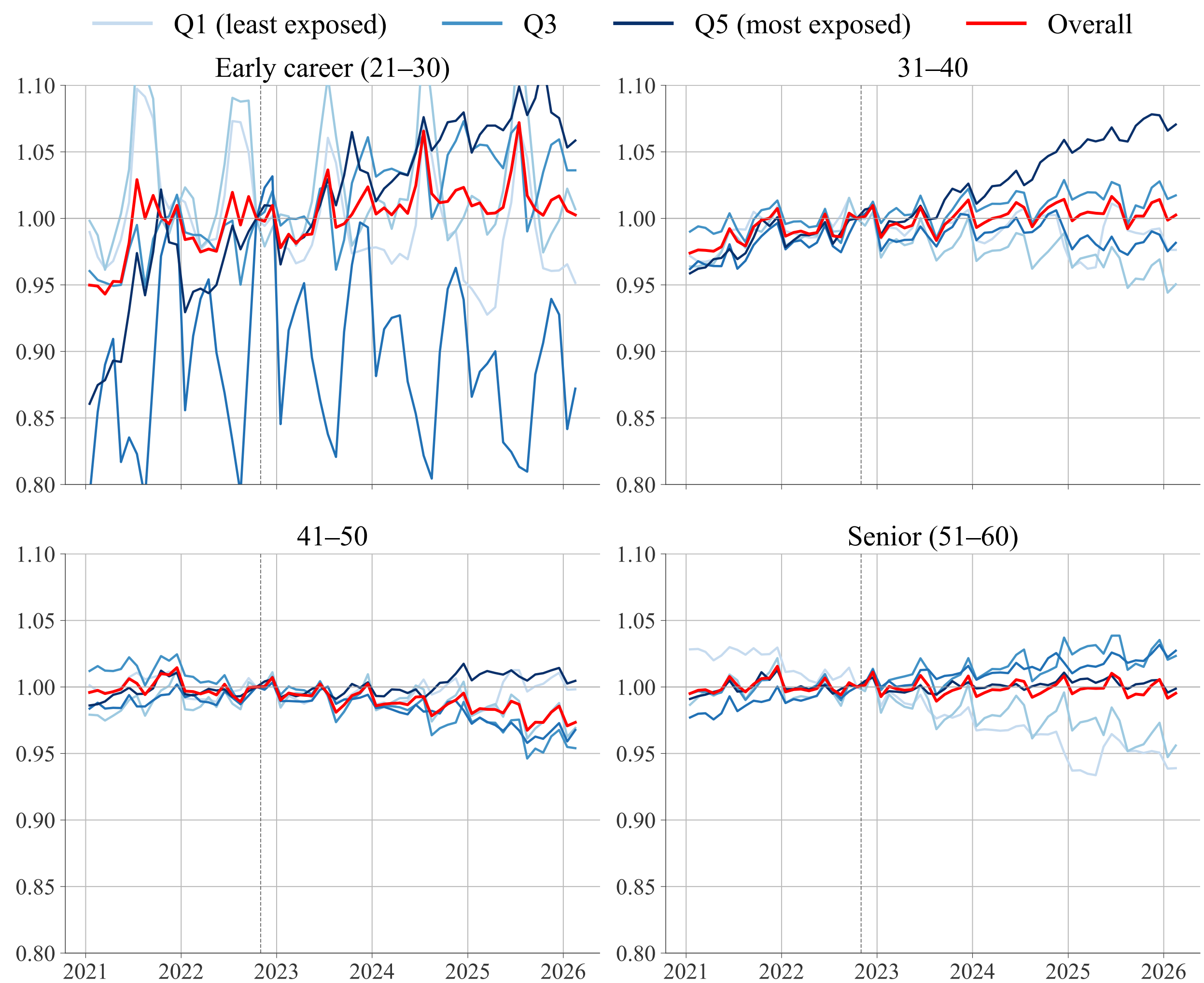

Figure A9: Employment by AI-exposure quintile and decade age group, public sector.
Notes: Public-sector employment per capita, indexed to October 2022 = 1. Occupations are ranked by the Eloundou et al. exposure score. Q1 (lightest) = least exposed; Q5 (darkest) = most exposed. Red line = overall age-group trend. Vertical dashed line marks the November 2022 ChatGPT release.